

Assignment "02"

Topic : "Mathematics In CS"

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Section : BCS-1B

Semester 1st

Activity "01"

❖ GRAPH IN FORM OF ADJACENCY MATRIX :

(i)

Points	0	1	2	3	4	5
0	0	0	1	3	0	0
1	0	0	0	5	0	0
2	1	0	0	2	1	4
3	3	5	2	0	7	0
4	0	0	1	7	0	2
5	0	0	4	0	2	0

(ii)

Points	A	B	C	D	E
A	0	1	7	4	3
B	1	0	6	2	5
C	7	6	0	4	5
D	4	2	4	0	1
E	3	5	5	1	0

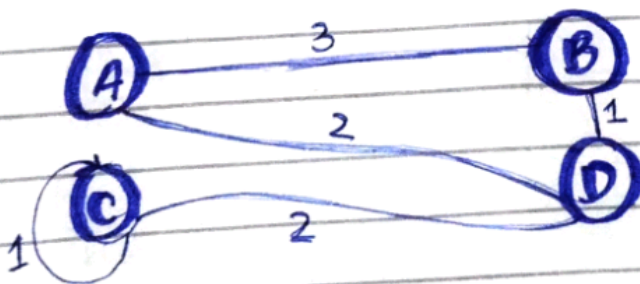
(iii)

Points	A	B	C	D	E	F	G
A	0	1	0	-2	0	0	0
B	-1	0	3	2	1	0	0
C	0	-3	0	1	4	0	0
D	2	-2	-1	0	2	0	-1
E	0	-1	-4	-2	0	3	0
F	0	0	0	0	-3	0	0
G	0	0	0	1	0	0	0

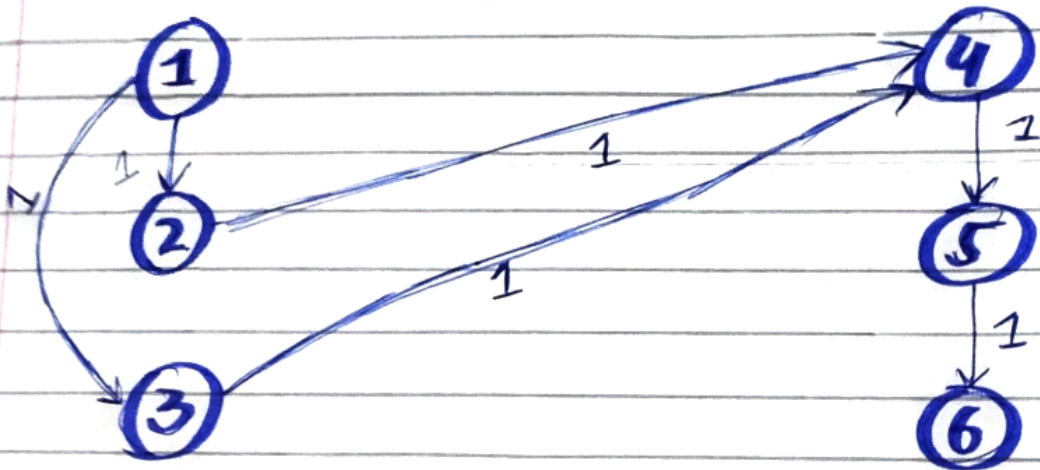
Activity "02"

❖ GRAPH OF ADJACENCY MATRIX :

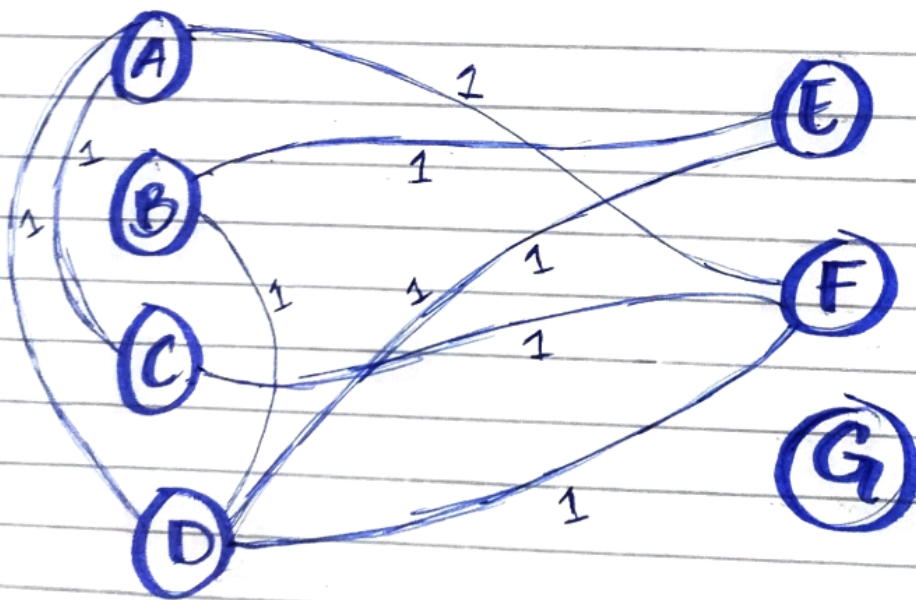
(i)



$(\overset{00}{11})$



$(\overset{000}{111})$

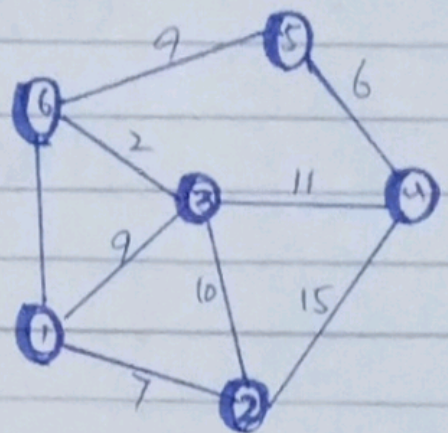


Activity "03"

❖ Find Shortest Path Between Two Nodes :

(●) Find Shortest path

The Shortest distance is "20"
which is from node
1 to 3 to 6 to 5 ($9+2+5$)



→ (●) Find Shortest path B/w node 1, 5

Activity "04"

❖ Shortest Path From Lahore To Gujrat

According to the given graph in assignment
the shortest distance from Lahore to Gujrat
is 44. i.e., Lahore to Kalashakaku to Guj-
ranwala to Sialkot to Gujrat ($14+18+6+6$)